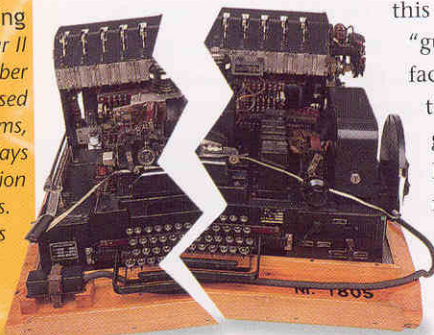


code-cracking
This World War II Geheimschreiber teleprinter used wheels, cams, cables, and relays to produce 2 billion possible settings. Its code was broken by the Swedish mathematician, Arne Beurling, who had access only to intercepted material. Today's methods of creating code would require quantum computers to crack them.



another, but in a quantum computer each guess (a string of 0s and 1s) would correspond to one of the superposed states, so the computer would make all the guesses and carry out all the calculations simultaneously. Shor showed that all the incorrect guesses would cancel each other out (like the destructive interference that makes dark stripes in the double-slit experiment), while the right answers would reinforce one another (like the constructive interference that makes bright stripes in the double-slit experiment), so that just one solution would emerge from the machine. All the wrong answers destroy one another.

permissible errors

A perfect quantum computer would be even more awesome. It would simultaneously solve every problem that it was capable of solving, and give the answer to every question that it was capable of answering, providing we knew how to program it and how to interpret the output. Such a prospect is a long way off, but in a sense it means that you only need to build one full-sized quantum computer and run it once, then spend generations trying to understand

this problem by “guessing” possible factors and multiplying them out to see if they give the extremely large number. This is more or less the way a conventional computer works, checking one possibility after

the answer is 42!

universal answer
A quantum computer the size of a laptop PC would be as powerful as a conventional computer containing every atom in the Universe. Our problem might be understanding the answer.



the output. But at the level we are likely to achieve in our own lifetimes, it wouldn't matter if the machine makes mistakes. A few of the quantum dots might, for example, fall out of their superposition of states due to the effects of random noise inside the machine and lead to a wrong

“Classical information is like the information in a book... the information in real microscopic physical systems is more like the information in a dream. It's certainly there in some sense, but if you try to tell somebody about your dream you won't remember it the same way it was before you told them.”

Charles Bennett, American physicist (2000)

answer. Such a machine would be so fast, though, that it would only need to produce the right answer once in a thousand times. To overcome this particular problem, you could run the computer a thousand times before breakfast

and then multiply all the possible answers on an ordinary computer to see which factors really do produce the extremely large number that you are interested in. Such an imperfect but very fast computer would be immensely valuable in the areas of military, political, and industrial espionage, and there is no doubt that the current funding that is available for such projects will continue. In this area of politically driven technological development, the less that is heard about it, the more likely it is that progress is being made. And once imperfect quantum computers exist, their reliability will improve inexorably. Then, their immense power might be able to help us turn other weird quantum possibilities into practical reality.

old-style espionage
Applying the power of a quantum computer to the discovery and decoding of secret information could make the role of the traditional spy obsolete.

